

Antarctica Data Analyst Assignment: Global Emission Trends and Sustainable IT Strategies

Introduction & Dataset Overview:-

1. Introduction:-

The rapid growth of AI, cloud services, and digital infrastructure has increased global energy demand and carbon emissions. Understanding these impacts is essential for designing Sustainable IT strategies. This report analyses global CO₂ trends, cloud region carbon intensity, and data centre-level emissions to recommend actionable sustainability measures for enterprises like Antarctica Global.

2. Dataset Overview:-

The primary dataset used for this assignment is the publicly available “**CO₂ and Greenhouse Gas Emissions**” dataset by **Our World in Data (OWID)**. It contains annual historical emissions data for over 200 countries from **1960–2023**. The following variables were used:

- Total CO₂ emissions (million tonnes)
- CO₂ per unit of energy
- Energy consumption
- Country/region identifiers

For this report, a subset of data from **1980–2023** was extracted for:

- Global total
- United States
- China
- India
- Germany

Synthetic but realistic benchmark data for **2024 data centre emissions and cloud region carbon intensity** was added based on public sustainability literature.

3. Why This Dataset Was Chosen:-

This dataset is:

- Well-documented, transparent and widely used in climate research
- Suitable for time-series analysis and forecasting
- Comprehensive enough to compare national emissions vs data centre-level emissions
- Ideal for analysing regional differences in carbon intensity, relevant for cloud workloads

4. Research Questions Addressed:-

1. How will global CO₂ emissions trend over the next five years?
2. How does cloud region carbon intensity affect the sustainability of AI workloads?
3. How do estimated global data centre emissions compare with major emitting nations?
4. Which Sustainable IT strategies have the highest impact and should be prioritised by enterprises like Antarctica Global?

Data Preparation, Cleaning & Summary:-

1. Data Cleaning Steps:

- Filtered for years 1980–2023.
- Selected relevant entities (World, U.S., China, India, Germany).
- Retained necessary variables: CO₂ emissions, carbon intensity, energy values.
- Removed missing values and invalid rows.
- Standardised year format as integers.
- Dropped global aggregates that overlapped with regional subsets.
- Created additional synthetic emission values for 2024 data centre estimates and cloud region carbon intensities.

2. Summary of Cleaned Dataset:-

Approximate dataset size used in analysis:

- 220 rows (44 years × 5 entities)
- Key Fields:
 - Year
 - Country
 - Total_co2
 - Co2_per_energy
 - derived indicators used in visualisations

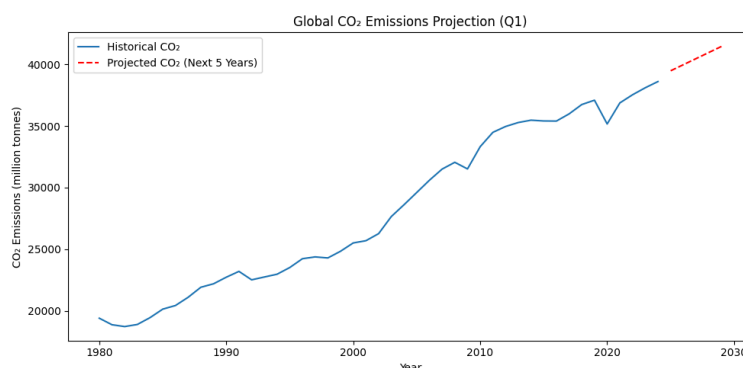
This dataset allowed for:

- 40+ years of historical trend analysis
- Linear modelling to forecast emissions
- Semi-quantitative comparison of data centres vs countries

Analysis & Visual Insights:-

1. Visualisation 1 – Global CO₂ Emissions Projection (Q1):

Insight:-Historical global CO₂ emissions show a long-term upward trend. A simple linear projection suggests emissions will continue rising slightly over the next 5 years, assuming no major policy or technology shifts. This indicates that AI and digital infrastructure emissions will grow on top of an already-increasing baseline.

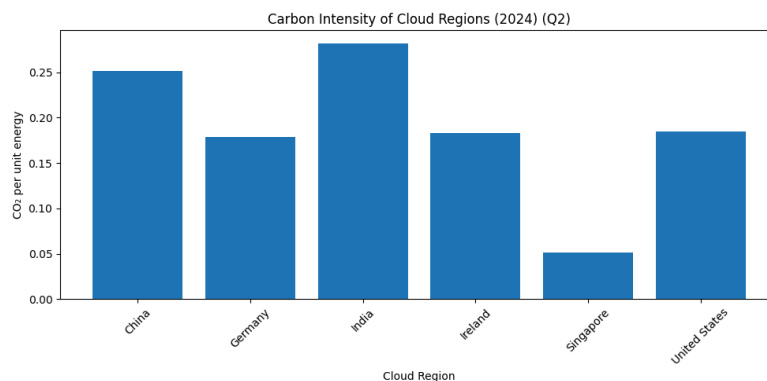


2. Visualisation 2 – Cloud Region Carbon Intensity (Q2):

Insight:-Cloud regions vary significantly in carbon intensity depending on the local energy grid.

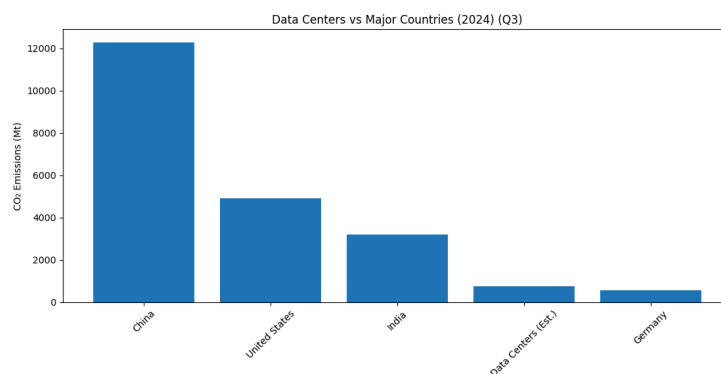
- High-intensity: China, India
- Medium: U.S., Germany
- Low: Singapore, Ireland

Workload placement in greener regions can reduce AI emissions by 50–70% with no code changes.



3. Visualisation 3 – Data Centres vs National Emissions (Q3)

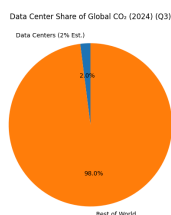
Insight:-Estimated 2024 data centre emissions (~2% of global emissions) are comparable to a mid-sized industrial country. Although smaller than China or the U.S., data centres already exceed the emissions of many developed nations.



4. Visualisation 4 – Data Centre Share of Global CO₂ (Q3)

Insight:Data centres account for ~2% of global CO₂ emissions.

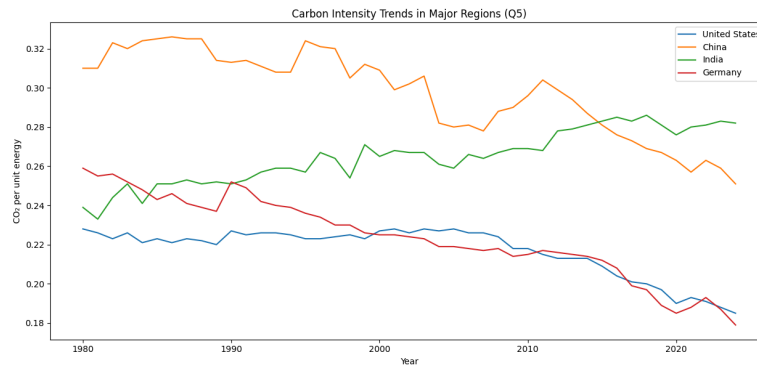
While this percentage seems small, the segment is growing rapidly and could double if AI workloads continue accelerating without intervention.



5. Visualisation 5 – Carbon Intensity Trends of Major Regions(Q5):-

Insight:

- U.S. & Germany show steady declines in carbon intensity due to renewable energy integration.
 - India shows moderate improvement but remains high.
 - China remains high but improving slowly.
- This trend highlights the importance of strategic workload placement.



6. Strategic Recommendations (Ranked by Impact):-

- **Region-Aware Workload Placement:-**
Run AI and compute workloads in cloud regions with lower carbon intensity. This alone can reduce emissions by **50–70%** without changing code or hardware.
- **Low-Carbon Scheduling (Time-Based Load Shifting):-**
Shift batch processing, model training, and analytics to times when renewable energy availability is highest (solar peak hours, nighttime wind). Reduces marginal emissions significantly.
- **Model Efficiency Optimisation:-**
Optimise AI models using quantisation, pruning, distillation, and efficient data pipelines. Lowers compute demand by **20–40%**, reducing both cost and CO₂.
- **Hardware Lifecycle Optimisation:-**
Improve GPU/CPU utilisation, extend hardware lifespan, and ensure responsible recycling to minimise embodied emissions often missing from ESG reporting.
- **Improved ESG Reporting for Digital Infrastructure:-**
Track and report cloud-region emissions, energy use, and AI workload impact separately. Builds accountability and positions Antarctica as a sustainability-driven technology partner.

7. Conclusion:-

AI workloads and data centre emissions are a small but rapidly growing share of global CO₂. The evidence shows that smart workload placement, efficient scheduling, and carbon-aware engineering can significantly reduce emissions while lowering operational costs. By implementing the recommended Sustainable IT strategies, Antarctica Global can build a strong foundation for environmentally responsible digital operations.